

From Tool Access to Research Capability: Tools and Memory for Financial Agents

Anonymous ACL submission

Draft under revision; not submission-ready.

Abstract

Financial research requires choosing appropriate methods, interpreting their outputs, and using experience to inform later decisions. How can a language-model agent turn access to professional tools into these capabilities? We study this question through FIN-SKILLS, a library of financial guidance and executable methods, and a framework connecting tool use with selective context and persistent memory. HiSTrim provides a mechanism for retaining historical context within a model, while an exploratory memory component consumes checked, delayed outcomes across episodes. Execution and feedback checks support these operations by recording what ran and which outcomes are eligible for learning. In an autonomous-use development study, two open-weight models completed 64 portfolio decisions and made 43 successful tool calls, but neither improved mean proxy return over its no-library control. The study therefore motivates examining how tools are selected and used, beyond counting successful calls. We relate these observations to separately reported recommendation results and specify comparisons of method selection, task quality, adaptation and computational cost. The combined financial tool-and-memory loop remains to be evaluated; the present evidence establishes autonomous use in the tested setting, not sustained improvement.

1 Introduction

A financial research agent must decide which method suits a task, what data that method requires, how to interpret its output, and when later evidence warrants changing its decision. For example, a forecast, a volatility estimate and a portfolio allocation answer different questions. An agent can execute all three correctly and still fail to connect them to the objective. Professional tool

access expands the available actions; research capability concerns how those actions are selected and used.

The distinction becomes more consequential across repeated decisions. A useful tool output can disappear from a limited context, while an earlier experience can continue to influence choices after conditions change. Within-task context selection and cross-episode memory address different parts of this problem. Their value depends on which information they preserve and how that information changes subsequent actions. An apparent success based on unavailable data is also a poor learning signal. Checks on execution and feedback help establish the conditions under which experience can inform a later decision.

We investigate these questions through FIN-SKILLS, which connects dated financial guidance to discoverable, executable methods. An agent can inspect method requirements, choose parameters, execute a tool, and use the returned result in its next decision. We relate this interface to two complementary memory mechanisms: HiSTrim selects historical representations within the model’s current context, and a fruit-fly-inspired memory component updates persistent state from delayed outcomes. The proposed connection concerns tool choice and decision adaptation. The current implementations and experiments cover individual parts of that connection.

Our completed autonomous-use study supplies a concrete starting observation. The tested models selected and executed tools, yet autonomous access did not improve mean proxy return over the no-library condition for either model. This observation does not identify a general failure of tool use or its cause. It motivates separating execution success from method appropriateness, interpretation and later adaptation. Differences in exposure and computation also matter when interpreting a portfolio outcome.

The paper asks three questions. First, how does professional tool access affect the methods an agent selects and the quality of its decisions? Second, what historical information should remain available under a context budget? Third, how should completed experiences influence subsequent choices as conditions change? Financial checks are supporting mechanisms for these questions; error reduction is one outcome to measure alongside useful completion, adaptation and cost.

This working draft contributes a reusable financial tool interface, a formulation connecting tool selection to within-task context and cross-episode memory, and a development study that measures actual model-selected calls and their decision outcomes. The study tests tool access. The recommendation evidence for HiSTrim and the implemented memory interface motivate further comparisons; they do not establish that the integrated financial agent learns better decisions.

2 Related work

FinAgent combines multimodal financial information with trading tools, reflection and memory (Zhang et al., 2024); StockBench evaluates sequential portfolio decisions using prices, fundamentals and news (Chen et al., 2025). The Finance Agent Benchmark evaluates financial research tasks (Bigéard et al., 2025). These works place tool use and memory within a broader study of language-model agents. Our development experiment separates fixed guidance, predetermined tool outputs and model-selected calls, providing a way to examine how a reusable library is used. It is not a matched leaderboard comparison with those systems.

Behavioral testing offers a complementary view of capability beyond an aggregate score (Ribeiro et al., 2020). Financial backtest overfitting and selection bias (Bailey and López de Prado, 2014; Bailey et al., 2017), and information leakage in financial agents (Li et al., 2025), motivate checks on the evidence used for decisions and learning. The persistent memory component draws on a model of interacting short- and long-term memory units (Huang et al., 2024). Within that component these are learned dynamics; they should not be confused with the separate distinction between a language model’s current context and memory carried across financial episodes.

3 Research capability across tools and memory

We use research capability to mean selecting methods suited to a task, producing useful decisions or analyses, and adapting later choices using available experience. This is an evaluation target, not a property inferred from the number of tools or successful calls. The framework separates the available methods, information retained for the current task, and state carried across episodes.

A decision and learning cycle. Let q_t describe the task, $\mathcal{D}_{\leq t}$ the records available by time t , and M_t the persistent state. A context mechanism selects or encodes information from the task history. A policy uses that context and a memory readout to choose a tool and its parameters, inspect its output, and eventually produce a decision. Once the episode ends, an eligible outcome may update M_t . These are conceptual interfaces for the combined agent; the current studies implement and test subsets.

Two distinct forms of memory. HiSTrim changes which historical representations survive within a model’s attention computation. Persistent memory changes learned state across episodes. Both can affect future choices, but they require different interventions: a context ablation changes what the current decision can attend to, while a memory ablation changes what previous outcomes contribute. Neither should silently alter the information cutoff or available tool catalog.

Experience as a connection to later action. The combined design would associate task conditions and tool outputs with a completed outcome, then expose a memory readout at a later choice. An update counts as useful only through its effect on subsequent tasks. A valid loss can be informative; a gain based on unavailable information should not be treated as successful experience. Checks test selected conditions for using feedback, while decision quality remains an independent evaluation outcome.

Original records remain available to the checking infrastructure even if a context selector omits them. HiSTrim requires access to a backbone’s internal representations and KV state, whereas the numerical memory adapter has a separate backend. Connecting both to LLM tool selection requires an explicit implementation and common task suite.

Component	Role in the research process	Capability question	Present evidence
FIN-SKILLS	Expose guidance, method requirements and executable tools	Which methods does the agent choose, and how does it use their outputs?	Financial library and autonomous-use development study
HiSTrim	Select historical representations while protecting task constraints	Which context can be removed while preserving decision quality?	Reported recommendation comparisons; financial integration untested
Persistent memory	Read and update state using completed experiences	Does experience change subsequent choices and improve adaptation?	Numerical adapter and feedback interface; combined policy untested
Execution and feedback checks	Record actual execution and test feedback eligibility	Which outputs and outcomes can support analysis or an update?	Regression checks and an audit feasibility study

Table 1: Roles and evidence within the proposed financial-agent framework. The rows describe complementary components, not an already evaluated integrated system.

4 Professional tools and domain knowledge

FIN-SKILLS exposes financial methods together with information an agent needs to use them: task scope, input requirements, dependencies, parameter conventions and dated source checks. Domain skills route requests to relevant libraries; per-library skills provide more detailed guidance. The algorithm catalog covers portfolio construction, forecasting, volatility and risk estimation, technical signals and execution schedules. It packages existing methods rather than introducing a new forecasting algorithm.

Agent-selected execution. An agent can discover a method, inspect its requirements and execute it with its own parameters. Structured outputs distinguish a numerical result from an unavailable backend or failed call. The agent then chooses whether to use that result, seek more information or finalize its action. An optional deterministic adviser supplies an additional suggestion; it does not replace the model’s choice. The autonomous study offers a price-compatible subset of the catalog and records requests and results separately from explanations.

Information and model interfaces. Retrieval interfaces expose relevant skill passages or caller-provided documents; collection records retain publication, observation and revision information when available. Availability cutoffs constrain the records exposed at a decision time. Forecasting and prediction adapters support fitting, prediction and persistence, while portfolio optimizers use execution interfaces suited to their inputs. These facilities support composing an analysis from multiple methods. The retrieval and collection additions

were not part of the frozen autonomous-use experiment. Appendix C records the interface inventory, implementation scope and version distinctions.

5 Within-task context selection with HiSTrim

A tool-using agent accumulates observations and intermediate outputs. Context selection determines which remain available for a later choice. HiSTrim provides a mechanism for selecting historical representations while retaining task constraints. The source study uses chronological interaction items for recommendation. A financial application would instead need to define meaningful record spans and preserve timestamps, asset identifiers and task requirements. This mapping is a proposed application; it is not part of the completed financial runs.

5.1 Protected, sign-preserving selection

Let $\mathcal{H}_{\ell-1}$ be the historical items surviving before pruning layer ℓ , and let $\mathcal{P}$ be a disjoint set of protected token positions. For an item representation h_m , the dual router computes

$$g_m = w_g^\top h_m + b_g, \quad u_m = w_u^\top h_m + b_u, \quad (1)$$

$$r_m = \sigma(g_m)u_m, \quad \sigma(g) = (1 + e^{-g})^{-1}. \quad (2)$$

For finite real logits, $\sigma(g_m) > 0$, so $\text{sign}(r_m) = \text{sign}(u_m)$; holding g_m fixed gives $\partial r_m / \partial u_m = \sigma(g_m) > 0$. By contrast, $\text{SiLU}(g_m)u_m$ is positive when both logits are negative. The sigmoid form removes this algebraic sign reversal. It does not prove that the utility score is correct or that thresholding will retain every useful item.

The router standardizes scores within a se-

quence and applies a learned threshold:

$$z_m = (r_m - \mu_\ell) / (s_\ell + \epsilon), \quad (3)$$

$$\mathcal{H}_\ell = \{m \in \mathcal{H}_{\ell-1} : z_m \geq \tau_\ell\}, \quad (4)$$

$$\mathcal{T}_\ell = \mathcal{P} \cup \bigcup_{m \in \mathcal{H}_\ell} \mathcal{I}_m. \quad (5)$$

Here $\mathcal{I}_m$ contains item m 's token positions. This separates item indices from token indices and ensures $\mathcal{H}_\ell \subseteq \mathcal{H}_{\ell-1}$. Protected positions remain in $\mathcal{T}_\ell$. Their retention prevents literal deletion, but changes in surrounding history can still change their hidden states and the model's answer. No equality of mutual information follows from masking alone.

5.2 Router fitting and physical compaction

The source method initializes routers from ridge-regression directions fitted on calibration data, normalizes each direction, and normalizes their weighted average. For nonzero fitted directions v_s , nonnegative weights summing to one, and a nonzero average, this gives

$$w_0 = \frac{\sum_s \alpha_s v_s / \|v_s\|_2}{\|\sum_s \alpha_s v_s / \|v_s\|_2\|_2}. \quad (6)$$

This construction fixes the norm; variance reduction needs additional distributional assumptions. A zero direction or cancelling average requires explicit handling. Forward-only refinement evaluates paired perturbations and updates directions in the tangent space, using $g_\perp = (I - ww^\top)g$ for unit w , followed by normalization. Thresholds are updated separately. These operations describe an optimization procedure, not a guarantee of convergence or avoidance of threshold collapse. Fixed numerical stabilizers also limit exact scale invariance of standardized scores.

At inference, HiSTrim-VarLen gathers retained entries into packed buffers with sequence offsets. Physical gathering distinguishes a smaller cache from a mask that leaves all entries allocated. Real memory and latency depend on the backend, sequence lengths, beam sharing, packing overhead and retained fraction; they must be measured for the model executing the financial task.

5.3 Attention compensation and its limits

Eviction changes attention normalization. For a fixed query, let s_0 be a sink logit and let $\mathcal{E}$ contain the removed positions, excluding the sink. Replacing its exponential mass by the sink plus the

removed mass would require

$$\Delta^* = \log \left(1 + \sum_{j \in \mathcal{E}} e^{s_j - s_0} \right). \quad (7)$$

Then $e^{s_0 + \Delta^*} = e^{s_0} + \sum_{j \in \mathcal{E}} e^{s_j}$ exactly for these fixed logits. Even this identity preserves only the denominator: preserving the attention output also requires the corresponding value-weighted contribution.

The source's practical offset $\beta \log(|\mathcal{E}| + 1)$ approximates the missing mass using only a count. Equality with Equation 7 requires a specific relation between removed logits and the sink; it is not an unconditional conservation theorem. We therefore retain it as an attention-compensation heuristic whose effect must be measured, together with protected spans and sign-preserving routing. Its utility is therefore evaluated through task quality and measured resource use.

Evidence and transfer. The recommendation study separates semantic-ID generation, rating prediction and hybrid ranking. Appendix A preserves its reported results and their provenance. Teacher agreement and label accuracy answer different questions; neither directly measures financial research correctness. A joint test must compare the same financial model with and without context selection under matched information and audit access, rather than transfer a recommendation gain to a return claim.

6 Cross-episode experience and persistent memory

Persistent memory changes what a decision maker carries from one episode to the next. The library exposes an exploratory fruit-fly-inspired component for studying how delayed outcomes can update that state. Its research question is whether memory operating over different timescales, updated from checked delayed outcomes, can support decisions to enter, persist, exit, and revisit opportunities under changing conditions and transaction costs. These policy capabilities remain development objectives.

Memory component. Our biological starting point is the mushroom-body memory model of Huang et al. (2024), which studies interactions between short- and long-term memory units. The

repository separates a numerical reproduction harness from an online adaptation. The adaptation advances its retention clock using observed conditioning events and reads memory without changing its state. Mapping signed financial returns to reinforcement and interpreting neuronal readouts as preferences are engineering choices. They are not validated biological correspondences, and the readout is not a calibrated expected return. The reproduction and derived model code are distributed as a separately installed GPL-3.0-or-later extension. The core library provides its adapter and model-card interface without bundling the extension or circuit parameters.

Connection to the library. The implemented feedback interface treats an entry and its subsequent holding periods as one completed decision episode. Given positive, cost-inclusive account values $W_0, \dots, W_T$, its learning target is

$$G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t) = \log(W_T/W_0). \quad (8)$$

This undiscounted complete-episode target includes entry costs and later holding outcomes. It requires the episode to end before the result can be used. The interface checks interval continuity, return accounting, and availability timestamps, and invokes the library’s source-lineage guard. Pending feedback waits; invalid feedback is rejected; valid negative returns remain learning signals. A completed receipt can be consumed once. The source account values still require independent execution reconciliation: these checks cannot establish that supplied values are authentic.

Current evidence and remaining comparison. Earlier custom sparse controllers made market decisions, but their development results did not establish a consistent financial advantage. Architecture diagnostics identified a mismatch between one-interval rewards and persistent holdings, return clipping that can change preference ordering, and impaired adaptation under repeated reward reversals. Those findings concern the tested implementations, not the suitability of mushroom-body models in general. The newer memory interface uses supplied outcomes in controlled learning cases; it does not yet autonomously choose entry, holding duration, or exit. The LLM tool-use study in Section 9 does not use this memory adapter.

The compact-memory benchmark now includes a frozen-memory control with the same volatility gate as the learned-memory arm, using identical data, costs, chronological splits and seeds. This separates the contribution of learning from that of the risk gate. No result for the newly added control has been recorded. The volatility gate controls exposure; it is distinct from the delayed-feedback checks described above.

The proposed comparison will hold information, actions, costs, and learning targets constant across controllers with and without feedback checks. Linear and dense controllers, non-biological models with matched memory timescales, and ablations of memory interactions are needed to distinguish the contributions of feedback validation and memory structure. Evaluation must measure adaptation, exit behavior, turnover, and net performance on untouched periods. These comparisons would test which retained experience changes later decisions and whether that change improves adaptation. Connecting the numerical readout to LLM tool choice remains a separate integration step.

7 Checks supporting execution and experience

Tool use and learning need records of what happened. The execution controller binds checks to the submitted code, inputs and numerical claims, recording their digests and results. Changed artifacts invalidate earlier receipts. A required-check policy reports `PASS` when all required checks ran and passed, `FAIL` for a detected defect, and otherwise `INCOMPLETE` for missing or rejected evidence. This mechanism establishes the scope of execution evidence; it does not choose the best method for the task.

For learning, availability and accounting checks constrain which delayed outcomes can be consumed. Pending feedback waits and invalid feedback is rejected; valid negative outcomes remain eligible. The purpose is to support interpretable experience updates, not to retain only profitable episodes. Passing these checks does not prove that supplied records are authentic or that the original choice was optimal.

The supporting audit study contrasts guidance, optional checking and required final checking. Its accepted model submissions were not independently gradable, so it does not establish an im-

provement in decision quality. Appendix B retains the protocol, historical pilot, instrument failures and regression evidence. Those records define the current checking scope separately from the capability study.

8 Evaluating use, retention and adaptation

The current development study measures autonomous tool use and portfolio outcomes. The broader framework requires comparisons at three stages. The following design specifies those comparisons; it does not report completed memory or context ablations.

Method selection and task quality. Tasks should expose method requirements and allow an agent to choose tools, parameters and stopping points. A method can be appropriate without being uniquely optimal. Selection can be assessed against task-specific eligibility and suitability criteria defined independently of the agent output, alongside the quality of the resulting forecast, allocation or analysis. Outcome-only rankings risk rewarding a lucky choice. Matched controls distinguish basic inputs, skill guidance, fixed tool outputs and agent-selected calls. A competing tool-access baseline is needed to isolate the library’s contribution from access to executable methods in general.

Context retention. With the backbone, available information and tools fixed, compare full context, simple budget-matched retention and HiSTrim. Measure task quality, retention of required evidence, allocated KV memory and wall-clock latency together. A smaller context or higher confidence alone is not an improvement. This comparison also requires defining financial record spans and protected task information before testing.

Experience and adaptation. With context and tool access fixed, compare no persistent memory, frozen memory, an updating non-biological baseline and the proposed memory component. Matched controllers should receive the same eligible feedback and action opportunities. Track how choices change after feedback, performance on later unexamined tasks, adaptation after shifts, turnover and cost. A separate feedback-check ablation tests update eligibility; a matched exposure control separates learning from a risk gate.

Context-by-memory comparisons are necessary to assess their interaction rather than attribute separate gains to the full system.

The primary measures should reflect the task: forecast accuracy or calibration, decision utility, or the supported quality of an analysis. Report valid completion, ineligible information use and execution failures as constraints on interpretation. Freeze outputs before independent scoring, preserve failed attempts, and use new tasks or chronological periods for evaluation. Report actual calls, tokens and latency, and compare matched budgets; portfolio claims additionally need cost and risk controls.

9 Development study: tool access and decision outcomes

This experiment examines the first part of the capability question: whether a fixed language model chooses professional tools and how access changes its portfolio decisions. HiSTrim and persistent memory are not used in these runs. An initial 12-trial pilot supplied predetermined tool results; it did not test autonomous selection. That distinction motivated the corrected intervention below. Both experiments use the same public market path, so the second is explicitly development evidence.

9.1 Matched controls and model-selected tools

We compare Qwen2.5-Coder 7B and 14B under four conditions: historical prices and basic summaries alone; the same inputs with fixed skill excerpts; excerpts plus predetermined tool outputs; and autonomous access to library tools. Each condition uses inference seeds 11 and 23 and the same maximum of five model responses per decision, each capped at 256 generated tokens. Conditions run in a frozen randomized order. Identical limits do not imply equal actual consumption, because the agent may finalize early.

The autonomous condition starts with tool descriptions. It can discover the algorithm catalog, read a related skill, run an algorithm with chosen parameters, or ask an optional heuristic strategy adviser about an asset. The model chooses whether to call a tool, which tool and parameters to use, and whether to make another call after inspecting the result. No particular call is forced. The dispatcher executes the requested API against

the current past-only snapshot and returns its actual output. Tool requests and results are recorded separately from model explanations; saying that a tool ran does not count as execution. The offered subset contains 18 price-compatible algorithms, not every capability in the library.

All conditions can finalize or use a response for further assessment. Invalid requests and formatting errors consume the existing budget and return feedback. A decision without a valid final allocation leaves existing holdings unchanged and remains a failure in the record; provider errors abort a run. Responses are saved before dispatch, and final allocations are frozen before independent scoring. This is a JSON action loop, not provider-native function calling or arbitrary generated-code execution.

9.2 Data, timing and scope

The existing ECB fixture supplies USD, JPY, GBP and CHF reference rates quoted per EUR. Their reciprocals represent foreign-currency values in EUR. These are indicative reference prices, not executable quotes; the ledger excludes interest and carry. Model messages anonymize asset identifiers and omit dates, which does not establish absence of pretraining contamination. No historical news archive is included.

Each decision sees 126 past observations. Evaluation covers the first 84 reference observations of 2025, ending May 2, with four decisions spaced 21 observations apart. Decisions after observation t execute at the reference price at $t + 1$ and cannot earn the preceding movement. Allocations are long-only, without leverage; residual weight remains zero-interest EUR cash. The primary cost is 5 basis points per side, including entry and terminal liquidation. Zero- and 20-basis-point scenarios rescore the same frozen decisions. Cash, equal-weight buy-and-hold and deterministic library rules are separate controls. Two inference seeds on one path are not two independent markets.

9.3 Completed runs and outcomes

The preliminary fixed-output pilot completed 12 trials and 48 valid decisions. At 5 basis points, mean full-context minus no-library proxy return was -0.455 percentage points for 7B and $+0.449$ points for 14B. This mixed result cannot answer the autonomous selection question.

The autonomous-loop implementation initially

Condition	7B	14B
No library	+0.100	-3.324
Skill text	-2.930	-4.567
Fixed tool outputs	-3.075	-3.030
Autonomous library	-3.984	-3.783

Table 2: Mean cumulative proxy return (%) across two inference seeds at 5 basis points. These are indicative FX price-only results, not executable investment returns. The window was inspected during the earlier pilot; this is development evidence.

rejected single Markdown fences around valid JSON, although the earlier final-action parser accepted them. We retained and cancelled that instrument-failure attempt, then corrected the parser consistently for all action types. The corrected run used unchanged prompts, models, seeds, tools and market window in a new workspace. Its instrument suite passed 26 tests covering input timing and action handling.

The corrected experiment completed all 16 trials and 64 final decisions. Autonomous 7B made 13 successful tool requests and autonomous 14B made 30, with no call or format errors in these conditions. Across all conditions, one unavailable-tool request was denied and three format errors occurred in the 7B text-only condition; the model recovered within its existing budget. Valid final decisions therefore do not imply error-free interaction traces.

The autonomous agents used different tools: 7B executed cross-sectional momentum, Bollinger reversion and volatility-targeted momentum, while 14B mostly executed Bollinger reversion and also read the trend-following skill. These observed choices establish that the model selected calls. They do not establish that selection was appropriate or optimal.

Table 2 reports every condition. Autonomous access underperformed the no-library condition by 4.084 percentage points for 7B and 0.459 points for 14B; it also underperformed fixed tool outputs by 0.909 and 0.753 points. Autonomous 14B outperformed its text-only control by 0.783 points. Thus successful tool execution did not translate into a mean advantage on either primary comparison in this development run.

Risk exposure differs materially: mean risky exposure for 7B was 0.500 without the library and 0.969 with autonomous access; for 14B it was 1.000 and 0.500, respectively. The results are

whole-policy comparisons, not risk-matched alpha estimates. Autonomous groups used 38 model calls per model across eight decisions, compared with 8 and 12 calls for the 7B and 14B no-library groups. The complete run consumed 156 calls, 1,155,782 prompt tokens and 10,165 completion tokens. Economic outcomes must be interpreted alongside this additional computation.

The same prompt, market window and seeds were retained after the parser correction; no unfavorable financial run was removed. Paired date-block intervals are descriptive for this short path, not confirmatory tests. A paper-level claim requires untouched evaluation periods, independent markets, executable-price conventions, risk-matched controls and a competing tool-access baseline. The result motivates studying the connection between method choice, output interpretation and later adaptation. It does not establish which of these explains the observed returns; that requires the matched comparisons in Section 8.

10 Discussion: from execution to useful experience

The development study distinguishes access, use and outcome. The agents did select and execute financial methods, but their additional calls did not improve the primary mean proxy-return comparison. Their tool choices, exposures and compute use differed. The records therefore motivate competing explanations: a method may be unsuitable, its output may be poorly used, or a reasonable decision may lose on the observed path. The present experiment does not distinguish these causes.

Selective context and persistent memory offer testable ways to change the information available for future choices. Context selection should preserve evidence needed for the current task at a lower resource cost. Persistent memory should influence later decisions through an explicit readout and update rule. A useful experience record would connect task conditions, selected tools, outputs and subsequent outcomes; whether such records improve later method choice is a hypothesis for the combined agent, not a result of the numerical memory interface alone.

Checks support this process by identifying unavailable inputs, incomplete execution and ineligible feedback. Reducing these defects can make a comparison interpretable, but professional capability also requires completing useful tasks

and adapting appropriately. Conversely, an adverse outcome can still be valid experience. The learning interface retains valid negative outcomes rather than filtering them out.

11 Conclusion

Professional tools create opportunities for financial agents to extend their methods; context and persistent memory determine which information can guide their choices. This paper organizes those operations around method selection, decision quality and adaptation, with execution and feedback checks as supporting infrastructure. The completed development study demonstrates autonomous tool use without a mean return advantage in its tested setting. Establishing research capability beyond that result requires showing how retained information and experience change subsequent decisions under matched, previously unexamined conditions.

Limitations

The empirical findings come from a short, previously inspected reference-price path and two models in one family. Inference seeds do not supply independent markets; reference prices exclude execution effects and carry. Actual compute and risk exposure differ across conditions. These observations cannot establish a general benefit or failure of professional tool access.

HiSTrim’s recommendation tables are author-reported, with the underlying training records and predictions still requiring verification. The financial mapping of its item spans and KV access is untested. The persistent memory adapter does not yet implement an autonomous entry, holding and exit policy or an LLM tool-selection update. Joint integration, cross-task transfer and sustained improvement remain open.

The audit feasibility study did not yield independently gradable model submissions. Finite checks can miss dependencies, and host confinement alone does not establish scorer isolation. Publicly inspected tasks and seeds are development data. Full protocols, failures and evidence qualifications remain in the appendices; none of these instrument checks substitutes for a capability evaluation.

Authorship, data and disclosure status

Author order, affiliations, funding, data permissions and final declarations require confirmation by the accountable authors. AI assistance was used for code review, manuscript revisions and experiment preparation; its scope and the human review record must be completed before submission. The manuscript remains a working draft.

References

- David H Bailey, Jonathan Borwein, Marcos López de Prado, and Qiji Jim Zhu. 2017. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69.
- David H Bailey and Marcos López de Prado. 2014. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94–107.
- Antoine Bigeard, Langston Nashold, Rayan Krishnan, and Shirley Wu. 2025. Finance agent benchmark: Benchmarking llms on real-world financial research tasks. *arXiv preprint arXiv:2508.00828*.
- Yanxu Chen, Zijun Yao, Yantao Liu, Amy Xin, Jin Ye, Jianing Yu, Lei Hou, and Juanzi Li. 2025. StockBench: Can LLM agents trade stocks profitably in real-world markets? *arXiv preprint arXiv:2510.02209*.
- Cheng Huang, Junjie Luo, Seung Je Woo, Lucas A. Roitman, Jizhou Li, Vincent A. Pieribone, Madhuvanthi Kannan, Ganesh Vasan, and Mark J. Schnitzer. 2024. Dopamine-mediated interactions between short- and long-term memory dynamics. *Nature*, 634:1141–1149.
- Xiangyu Li, Yawen Zeng, Xiaofen Xing, Jin Xu, and Xiangmin Xu. 2025. Profit mirage: Revisiting information leakage in llm-based financial agents. *arXiv preprint arXiv:2510.07920*.
- Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with Check-List. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4902–4912.
- Wentao Zhang, Lingxuan Zhao, Haochong Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, Longtao Zheng, Xinrun Wang, and Bo An. 2024. A multi-modal foundation agent for financial trading: Tool-augmented, diversified, and generalist. *arXiv preprint arXiv:2402.18485*.

A Recommendation evidence retained from the source manuscript

This section preserves the substantive HiSTrim comparison from the shared manuscript. The source is the Overleaf revision uploaded by Shwai He on September 22, 2026, with the interface displaying 19:05. Its TeX and figures were downloaded before merging. The tables below are author-reported manuscript values, not an independent replication: the original training code, user-level predictions and run receipts were not available in this workspace. The local merge record stores source hashes and arithmetic checks. This evidence status is separate from the financial result files used elsewhere.

Generative recommendation and baselines.

Table 3 retains all five task columns and all six method rows per model scale from the source’s principal generative table. Keeping every row shows where other methods outperform HiSTrim, including the Video task. The table does not support universal superiority across tasks. Reported KV retention also differs between methods, so these rows are not perfectly matched resource budgets.

Calibration-seed variability. The source appendix lists five calibration partitions for OneRec-1.7B Ad. Recomputing from its rounded rows yields Pass@32 of $18.78\% \pm 0.72\%$ and MRR@32 multiplied by 100 of 3.81 ± 0.19 (mean and sample standard deviation). The Pass@32 mean resolves the source main text’s differing rounded value. These are variations across fitted routers evaluated on the stated test set, not independent test populations. Significance claims require the original paired predictions and sampling procedure and are not inferred from these summary rows.

Task accuracy, agreement and cost. Table 4 retains both favorable comparisons and a counterexample to treating teacher agreement as task success. On Beauty, the source reports lower KL for HiSTrim than for TokenDrop-50, while its label MAE is higher. On the hybrid ranking task, the aggressive variant is faster but has lower NDCG than the dense baseline. The Goodreads comparison reports lower MAE with reduced KV; the original prediction records remain necessary to assess uncertainty. These contrasts motivate joint measurement of quality and resources rather than a single aggregate claim of improvement.

Items requiring reconciliation. The source’s figure captions and tables contain different aggregate configurations; we retain the explicit table rows rather than combining their maxima. The source also describes several calibration-set sizes without a complete run-to-configuration map. Hardware timings, significance markers, broad baseline compatibility claims and the proposed transfer to financial agents require their supporting run records. The untouched source remains available in the pre-merge backup. These gaps do not alter the financial study’s recorded outcomes and are not filled with new experiments or assumed results in this merge.

Model	Method	Hist. KV (%)	Ad	Product	Video	Label-cond.	Interactive
1.7B	Full Dense	100.0	3.24	7.81	5.16	1.25	11.95
1.7B	TokenDrop-50	50.0	3.08	7.72	4.89	1.29	11.70
1.7B	H2O	54.6	3.98	7.51	7.77	1.25	11.72
1.7B	SnapKV	54.6	3.85	7.75	7.36	1.30	11.85
1.7B	Router-Tuning	45.7	4.00	7.27	3.30	1.18	11.24
1.7B	HiSTrim	47.5	4.10	8.09	6.43	1.50	12.08
8B	Full Dense	100.0	4.56	8.22	7.67	1.41	18.25
8B	TokenDrop-50	50.0	4.22	7.89	7.12	1.32	17.40
8B	H2O	54.6	5.06	8.28	8.94	1.37	18.95
8B	SnapKV	54.6	5.02	8.25	8.54	1.72	18.70
8B	Router-Tuning	46.2	4.12	7.95	6.84	1.34	17.65
8B	HiSTrim	49.5	5.25	8.54	7.96	1.47	19.21

Table 3: Author-reported generative recommendation results retained from the Overleaf source table: MRR@32 multiplied by 100, with the source reporting 1,000 users per task and 32 beams. Historical KV retention excludes protected tokens. These manuscript values have not been reproduced from predictions; source significance markers are omitted pending the underlying paired records.

Task / model	Method	Hist. KV (%)	Label metric	Teacher metric	Speedup
Beauty / Qwen3-0.6B	Dense	100.0	MAE 1.642	KL 0.000	—
	TokenDrop-50	50.0	MAE 1.517	KL 0.233	—
	HiSTrim	50.1	MAE 1.643	KL 0.084	—
Goodreads / Qwen3-8B	Dense	100.0	MAE 1.232	—	—
	HiSTrim	35.0	MAE 1.078	Teacher MAE 0.651	—
Amazon / DCNv2+LM	Dense	100.0	NDCG@10 0.5183	—	1.00×
	TokenDrop-50	50.0	NDCG@10 0.5204	—	1.79×
	HiSTrim	50.0	NDCG@10 0.5567	—	1.78×
	HiSTrim aggressive	35.0	NDCG@10 0.5065	—	2.82×

Table 4: Selected author-reported rating and ranking comparisons from the source table. The source reports 357 Beauty, 250 Goodreads and 40 Amazon ranking cases. Lower MAE and KL are better; higher NDCG is better. Missing entries are not measured zeros. Teacher agreement, task-label performance and speed are kept in distinct columns.

B Supporting audit and validation studies

These studies characterize the execution and feedback checks used by the research framework. Their outcomes are distinct from method selection, decision quality and learning. They are retained to document what the checking infrastructure can measure and where the current instruments remain incomplete.

B.1 Independent evaluation

A seeded synthetic market provides quoted prices, corporate actions, listings, post-close commentary, precomputed scores and fundamental disclosures. A submitted function returns portfolio weights; a separate report states its net Sharpe ratio. The task now specifies that session- t weights must use information available by the end of session $t - 1$, consistent with close-to-close return accounting.

Historical arm	Mean absolute Sharpe gap
Opus, no library	0.188
Opus, library	0.034
Haiku, no library	0.574
Haiku, library	11.949

Table 5: Exploratory pilot: three runs per arm. Values come from saved grades; model versions and full interaction-level execution provenance require further verification.

The independent evaluator redraws future data and individual sessions, measures changes in submitted positions, checks post-delisting exposure, and recomputes performance at a fixed transaction-cost convention. Public intervention checks do not call this oracle. Private grades are generated only after the submission receipt is saved and are not returned to the agent for repair. Rejected submissions are also graded when executable.

The oracle’s fixed-cost Sharpe and the public tool’s stated-cost Sharpe answer different questions. Their discrepancy can reflect accounting conventions as well as coding errors. Neither should be called an estimate of genuine alpha without timing and other validity checks. A latent-drift strategy’s sample Sharpe is a reference observation, not a theoretical upper bound. Exceeding it is not mathematical proof of leakage.

B.2 Supporting validation records

Synthetic regression checks. Saved robustness results record 36 detections among 36 planted instances: twelve defect families repeated across three seeded worlds, with no reported clean-control false alarms in those worlds. This is regression evidence within one synthetic design. Repeating a seed design does not supply independently authored defects or estimate a general real-world false-positive rate. The detection suite exercises thirteen guards, not every registered guard.

Historical agent pilot. The saved pilot comprises three markets, two model tiers, and library-present versus library-absent conditions, with one run per cell. Table 5 recomputes the mean absolute discrepancy from the saved raw grades. The observed relative reduction for Opus is 81.9%, computed before rounding. It is not a significance claim.

The Haiku library arm has the largest discrepancy in this pilot. This motivates testing whether

guidance, tool use, or other differences explain the outcome. It does not show that documentation causes weaker models to fail or that mandatory checks solve the problem. Historical reports of unexecuted guard citations require original execution logs for verification; the saved grade table alone does not establish that behavior.

Numerical and real-data checks. The numerical benchmark compares implementations with reference calculations using case-specific tolerances. Passing within tolerance is not exact equality. Runtime measurements report the names and number of guards actually exercised, skipped and rejected checks, and machine-dependent timing. A fitted slope across a few data sizes is descriptive, not an asymptotic complexity bound.

The KOL fixture supports recomputing profile-group aggregates. Its model prediction metrics are saved summaries, not values independently recomputed from predictions and outcomes. In particular, its stored pooled Rank IC and mean daily cross-sectional Rank IC can have different signs. We do not treat this fixture as end-to-end reproduction, evidence of profitable trading, or external validation of the agent intervention. A separate evaluator now requires timestamped row-level predictions and outcomes, reports both forms of IC, and uses paired date-block uncertainty. Those source records remain an outstanding evidence requirement.

B.3 Four-condition audit study

The new protocol compares task guidance with ordinary accounting feedback, the same setup plus a fixed skill excerpt, the excerpt plus optional executable guards, and the excerpt plus a mandatory final audit. Market data, inference seeds and budgets are paired. Every condition has the same turn and response-token caps; actual token use, execution time and tool calls are also recorded. Conditions run in a frozen randomized order. Failed requests have no hidden retry or synthetic response fallback.

The primary outcome of this audit study is incorrect accepted submissions divided by all attempted submissions. Correctness among accepted outputs and overall acceptance rate must be reported alongside it, so refusing every output cannot appear to solve the task. Secondary outcomes include independent leakage measures, reporting discrepancy, post-delisting exposure, un-

executed and stale citations, repair success and resource cost. Missing, rejected, ungradable and provider-failed runs remain separate outcomes.

The first Beacon run is a public-seed feasibility study using pinned open-weight models. It tests operation and supports cost and sample-size planning. A confirmatory study additionally needs independent defect authorship, retained clean cases, repeated runs, and verified scorer isolation. Market-level paired uncertainty must preserve repetitions within markets. The present protocol is an amendment after examining the pilot, not a preregistration of its observations. Failure to find a return difference would not establish equivalence.

Operational findings. The first Beacon pre-flight rejected valid tool-call JSON wrapped in Markdown fences. We archived those records and amended the parser before a separate batch; this was an interface failure, not evidence of financial reasoning failure. In the revised batch, Qwen2.5-Coder 7B and 14B each completed 12 recorded cells, with 0 and 7 accepted submissions respectively. Neither model produced an independent numerical grade: final code failed input or output validation, or a required artifact was absent. Accepted does not mean correct, and these observations cannot estimate improvement in financial integrity.

Table 6 shows the per-condition breakdown for 14B, the only model that produced accepted submissions. Acceptance rate was 67% under C0 and C1, 100% under C2 (optional guards), and 0% under C3 (enforced audit): all three C3 runs reached the turn limit without producing a passing submission. Mean token use under C3 was 47,491 versus 13,877 under C2, indicating substantially greater iteration before the turn limit was hit. All accepted submissions were ungradable; the primary outcome—incorrect accepted submissions per attempt—cannot be estimated from this batch.

The 32B run encountered nonfinite generation values and a CUDA error before a tool call. Subsequent failure records from that process are not independent capability observations. At the time of that feasibility report, a separately recorded single-GPU recovery run was queued. A handwritten causal control passed the mandatory audit and independent timing/accounting checks on one public market. That control verifies an executable path,

Model	Condition	Accepted/Planned	Mean tokens	Primary status
14B	C0 no library	2/3	13,535	1× turn limit
	C1 text only	2/3	13,095	1× turn limit
	C2 optional guards	3/3	13,877	all accepted
	C3 enforced audit	0/3	47,491	3× turn limit
7B	all (12 cells)	0/12	—	turn limit / error
32B	all (12 cells)	0/12	0	provider error

Table 6: Per-condition Beacon feasibility results. All accepted 14B submissions were ungradable; mean tokens are averaged over all completed cells in the condition. 32B token count is zero because the process failed before any tool call. Values come from `BEACON_FEASIBILITY_RESULTS.json`; numbers in `evidence_numbers.tex` are generated by `scripts/build_paper_evidence.py`.

not general scorer validity. Before a confirmatory study, basic artifact validity must be distinguished from the incremental contribution of financial checks.

Component	Entries
Skill documents	129
Algorithm-backend catalog	48
Registered research guards	36
JSON-callable tools	56

Table 7: Inventory of the frozen study snapshot. Algorithm entries can expose existing methods through different backends. The autonomous study offers a price-compatible subset, not every catalog entry. This snapshot predates the September 22 RAG and Jev additions described below.

C Library implementation and interface scope

Knowledge and executable interfaces. Domain skills route tasks to relevant libraries, while optional per-library skills describe conventions, dependencies, and failure modes in more detail. The Python package exposes catalog search, skill loading, and reference retrieval. Packaged skill documents and scripts are generated from the same repository sources. A verification date records a source check; it does not guarantee that a statement remains current. Table 7 describes the frozen study snapshot. These counts measure available interfaces, not independently validated capabilities.

Algorithm selection and execution. The algorithm catalog describes input requirements, optional dependencies, and supported tasks. Implemented families include portfolio construction, forecasting, volatility and risk estimation, technical signals, and execution schedules. Recommendation rules filter candidates by the request and data requirements, then rank compatible entries using explicit preferences. They are not a learned predictor of future returns. A catalog entry does not imply that its dependency is installed or that execution will succeed; unavailable backends and execution failures remain explicit. Automatic execution does not silently substitute another method after a selected method fails.

Reusable models and lifecycle. The library supports direct Python use as well as agent-facing discovery and execution. Forecasting, regression, and classification adapters expose a common fit-predict lifecycle, including ARIMA, Ridge, random forests, LightGBM, and XGBoost when their dependencies are installed. Fitted models can be saved and restored with recorded environ-

ment and input metadata. This loader restores trusted user-created artifacts; it is not a pretrained-checkpoint download service. Portfolio optimizers and rule-based signals use execution interfaces without pretending to share a supervised training lifecycle. The current model registry also exposes an experimental fruit-fly memory adapter through `create_model("fly_memory")`. Its numerical backend is the separately installed GPL-3.0-or-later `fin-skills-fly` extension and requires caller-supplied circuit parameters. It supports prediction, checked feedback updates and persistence; it does not implement a complete autonomous trading policy.

Built-in retrieval and model composition. The RAG interface chunks packaged skills or caller-supplied documents and retrieves passages using local BM25. Callers may supply embedding, reranking and text-generation functions. Passages retain their document identifiers, source offsets and availability timestamps; as-of retrieval excludes documents unavailable at the requested time. The index resides in memory and can be persisted as versioned JSON. Citation checks validate references to retrieved passage labels, not whether a passage supports a claim.

The shared model registry includes a hosted TypeSafe Jev adapter for structured decisions and passage reranking. It requires service access and explicit network permission; no Jev weights are distributed with the library. The adapter can serve as a RAG reranker while a caller-selected model generates the answer. These September 22 additions have interface tests, including mock Jev transports. Live-service inference, retrieval quality and reranking gains were not evaluated in the earlier studies. A paired development runner now freezes BM25 candidates, the time cutoff and context budget before Jev inference, with relevance scoring performed separately on saved outputs. Its public synthetic fixtures test this workflow. A quality comparison still requires independently annotated retrieval tasks and an unchanged answering model.

Information availability. News collection supports RSS/Atom feeds, static pages, GDELT, and disclosure sources. SQLite stores collection records, revisions, polling checkpoints and pending alerts; the operator explicitly starts collection. The package does not ship a populated financial dataset, and this record store is separate from the

RAG index. Records distinguish publication time from observation time and retain revision information when available. An as-of digest filters future, stale, and undated records and deduplicates canonical URLs. These mechanisms support timestamp-aware processing; they do not reconstruct an independently validated historical news archive. Keyword risk indicators are not calibrated sentiment estimates, and absent news is treated as unknown rather than evidence of low risk. News collection is not exercised in the portfolio experiment reported here.

Model choice and rule-based advice. An agent can read a skill, inspect available algorithms, execute a selected method with its own parameters, and use the returned result in its next decision. The library also provides an optional market-state adviser based on trailing price and volatility statistics. Its trend, range, and stress categories are rule outputs, not learned regime probabilities or validated predictions of which strategy will outperform. A suggested strategy is therefore additional evidence the model may inspect. The autonomous study records the model's actual calls; it does not equate access to this adviser with autonomous algorithm selection.

Research workflow and boundaries. Research interfaces support data preflight, candidate evaluation, chronological validation, holdout reporting, and trial records. Holding final data out of candidate ranking reduces one avenue for selection bias, but the caller must still provide valid timestamps and data provenance. Python and JSON tool interfaces expose structured results; guards requiring live Python objects are not all available as JSON calls. Neither a data record nor a successful tool call certifies the whole research process.